Postdoctoral Fellow in Termite Behavior and Evolution
Computational approaches | Collective behavior | Comparative biology
Mizumoto Lab, Auburn University

The Mizumoto Lab is recruiting a postdoctoral researcher to join an NSF-funded project on the evolution of termite collective behavior and nest building (NSF Award 2611846). The anticipated start is in 2027, with some flexibility in timing. The position is expected to be available for up to three years.

Research focus
The project asks how individual-level behavior gives rise to group-level patterns and how these processes have evolved across termite lineages, especially focusing on tunnel excavation and nest construction. The postdoc will use quantitative and computational approaches to study termite behavior and evolution, with the flexibility to align with their expertise and interests. Planned and ongoing projects include

- Image and video analysis to characterize tunneling behaviors across ~20 termite genera
- Develop data-driven models to connect individual- and group-level behaviors across species
- Phylogenetic comparative analysis to reconstruct the evolutionary history of termite social behaviors

These provide several starting points. The postdoc will have substantial intellectual freedom to choose appropriate analytical approaches, develop new questions, and shape the direction of the research.

Qualifications
Applicants should have a PhD by the start date, a research record, and an interest in animal behavior and evolution. We especially welcome researchers who use quantitative or computational approaches. Relevant experience may include image or video analysis, tracking and trajectory analysis, comparative methods, network analysis, mathematical modeling, agent-based simulation, or related approaches. Experience with termites or social insects is not required.

Application
Interested applicants should email Dr. Nobuaki Mizumoto (nzm0095@auburn.edu) with:

- CV including links to papers and/or preprints
- A brief statement describing your research goals and interests, relevant experience, and interest in the position

The first review of applications will begin on December 22, 2026. Applications received after this date will continue to be considered until the position is filled.

Auburn University is an R1 research university located in a vibrant college town with access to diverse natural areas. The Department of Entomology & Plant Pathology includes a collaborative community with several laboratories studying social insects. The selected candidate will also be required to complete Auburn University's formal application and hiring process.

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